

Einteilung Präsentation/Protokoll

Thema	Max. Studis	Studi 1	Studi 2
Genomische DNA a) Konzentrationsbestimmung b) RNase-Verdau c) Restriktionsenzyme	2	Peko	Sonnleitner
Plasmidisolierung a) unverdaut/1x verdaut/2x verdaut	2	Luftensteiner	Lummerstorfer
TAS2R38 & LAC	2	Bleier	Suess
Proteinbestimmung	1-2	Simon	Grösswang
HPLC	1	Haselberger	Zec
ELISA	1	-	-

Genomische DNA (& Plasmide)

BLT4UE

SS2026

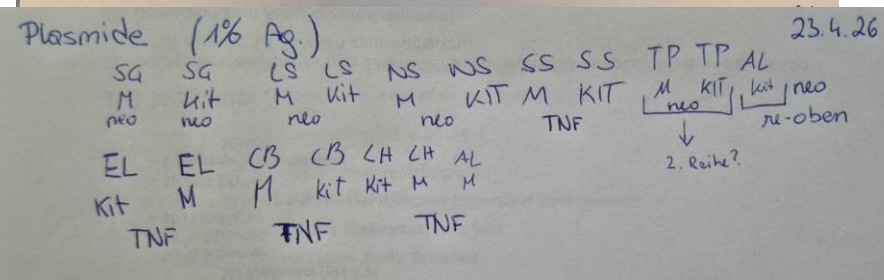
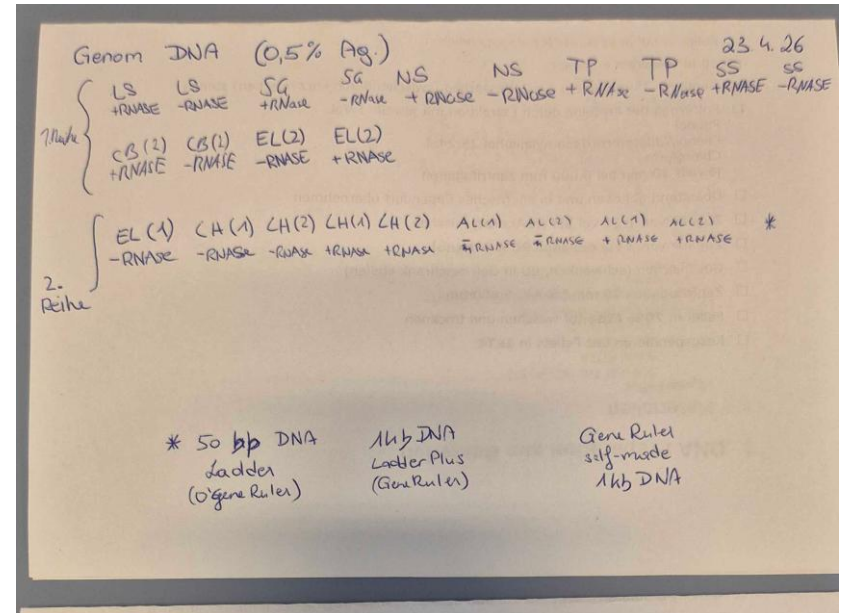
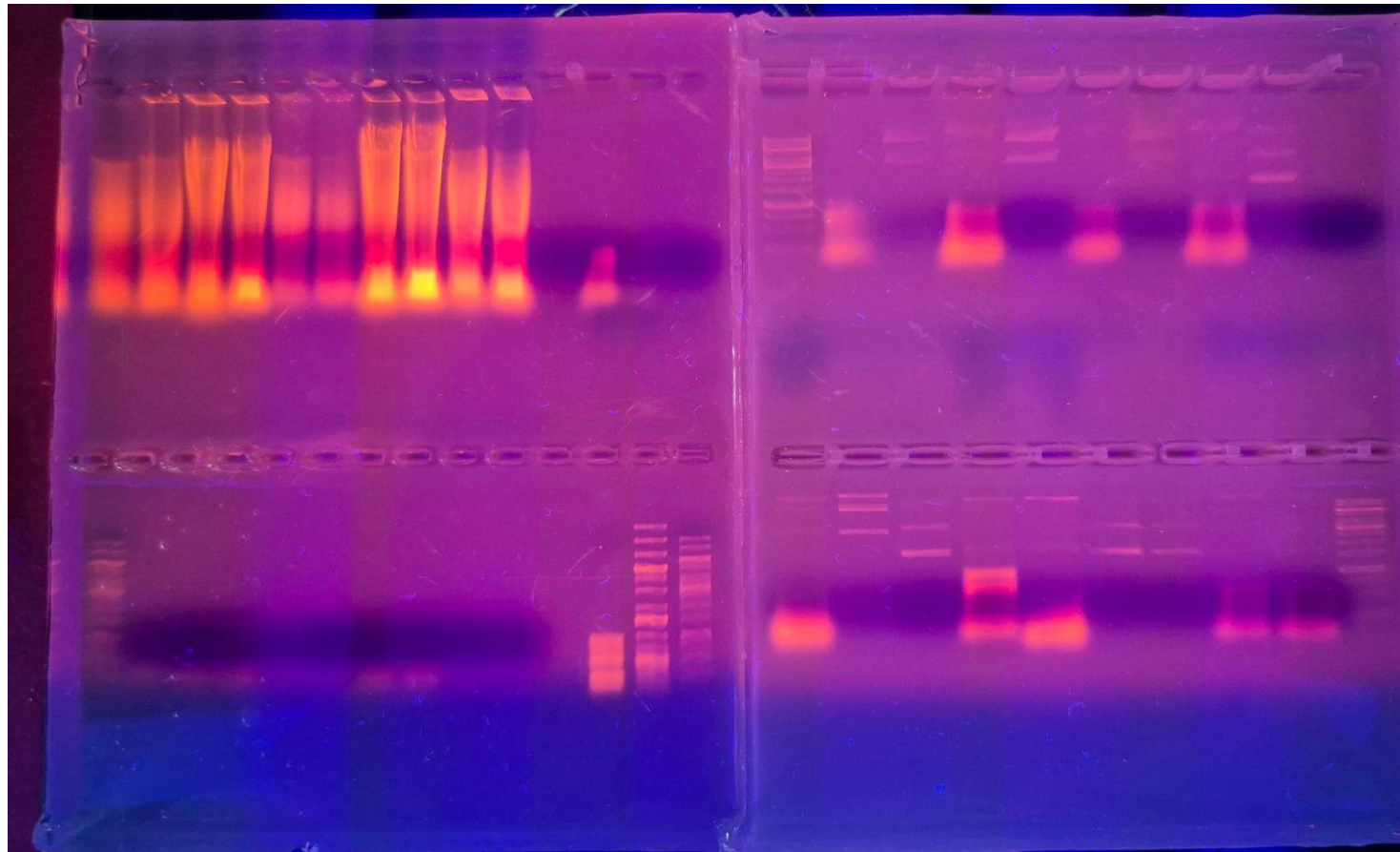
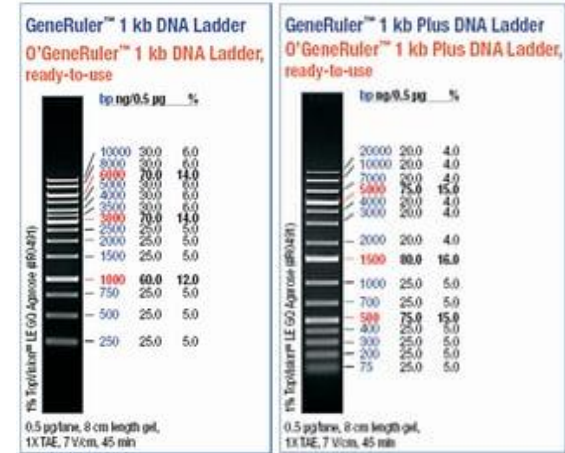
Gelelektrophorese

Genomische DNA

Schweineleber & bakterielle DNA

Plasmide

(Isolierung mit & ohne Kit)



Gewicht Leber

Leber
deonie : 3.92g
Tim: 3.48g
Natalie: 4,04g
Samuel: 3,71g
Sebastian: 4,43g

Photometrische Messung (Bakterielle DNA)

O=		RNase		ohne RNase		Bakterien	23.04.2026
		P1	P2	P1	P2		
230		-0,004 A (CB)	0,0064 (CB)	0,006 A (CB)	0,006 A (CB)		
		0,020 A (AL)	0,004 A (AL)	0,006 A (AL)	0,006 A (AL)		
		0,123 A (IZ)	0,015 A (IZ)	0,015 A (IZ)	0,021 A (IZ)		
		(EL)	-0,003 A (EL)	-0,008 A (EL)	-0,009 A (EL)		
		0,093 A (CH)	0,098 A (CH)	0,092 A (CH)	0,095 A (CH)		
260		0,019 A (CB)	0,040 A (CB)	0,040 A (CB)	0,040 A (CB)		
		0,023 A (AL)	0,013 A (AL)	0,010 A (AL)	0,040 A (AL)		
		0,140 A (IZ)	0,025 A (IZ)	0,037 A (IZ)	0,030 A (IZ)		
		(EL)	-0,000 (EL)	0,003 A (EL)	-0,002 A (EL)		
		0,093 A (CH)	0,092 A (CH)	0,091 A (CH)	0,096 A (CH)		
280		0,012 A (CB)	0,021 A (CB)	0,021 A (CB)	0,021 A (CB)		
		0,023 A (AL)	0,014 A (AL)	0,021 A (AL)	0,021 A (AL)		
		0,081 A (IZ)	0,019 A (IZ)	0,033 A (IZ)	0,033 A (IZ)		
		(EL)	0,007 (EL)	0,010 A (EL)	0,004 A (EL)		
		0,055 A (CH)	0,056 A (CH)	0,055 A (CH)	0,058 A (CH)		

Photometrische Messung (Leber)

23.4.26

Leber	+ RNase				- RNase			
	TP	NS	SG	LS	TP	NS	SG	LS
230	0.573 0.104	0.104	0.467	0.155 0.155	0.197	0.102	0.383	0.162 0.162
260	0.782 0.782	0.211	0.656	0.277 0.277	0.273	0.074	0.546	0.276 0.276
280	0.545 0.545	0.124	0.449	0.196 0.196	0.205	0.047	0.371 0.371	0.192 0.192
	Summe				Summe			
230	0.135				0.286			
260	0.272				0.408			
280	0.136				2.46 0.264			

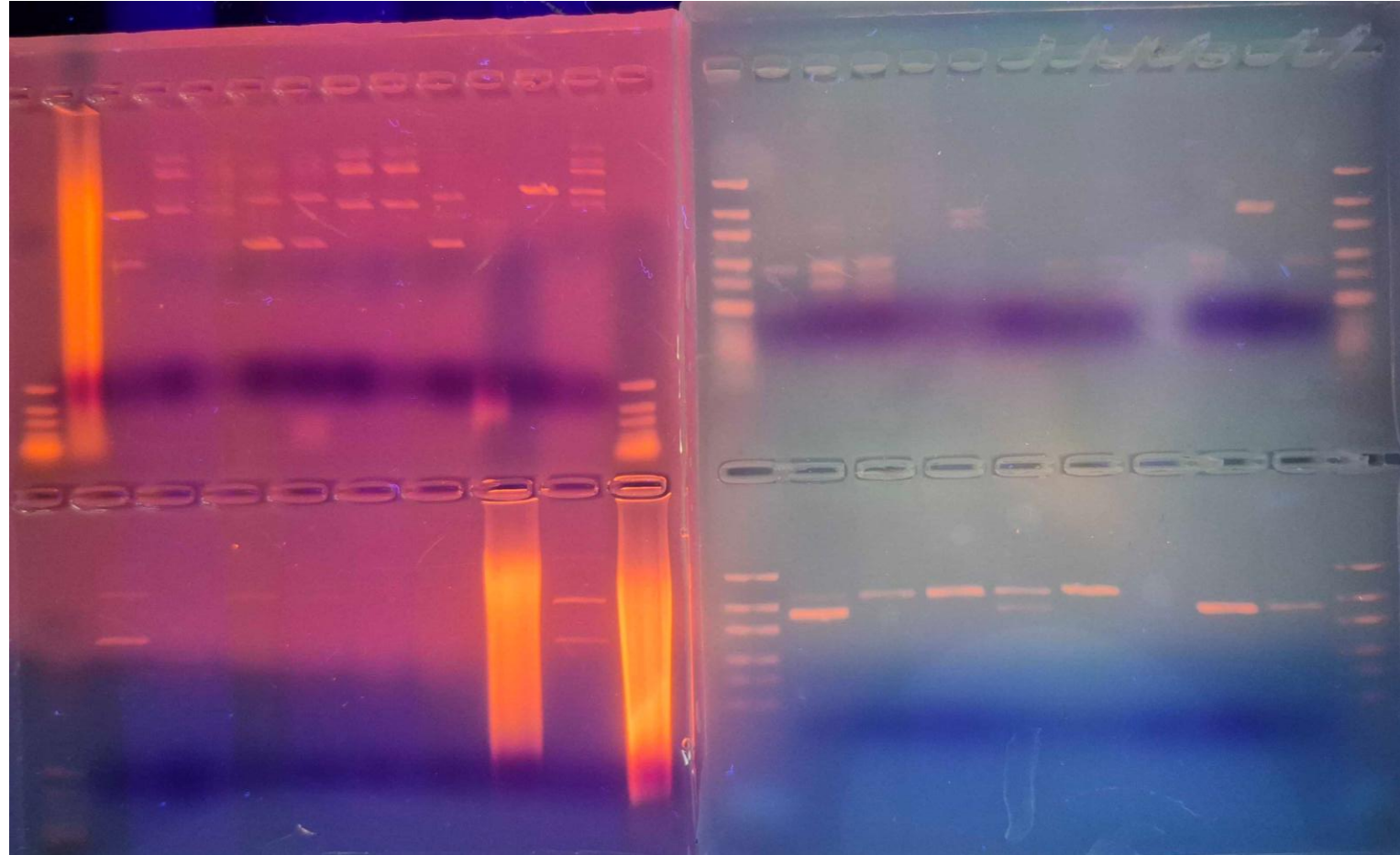
TAS

LAC

Restriktionsenzyme
(genomische DNA & Plasmide)

Überblick Gel 28.05.2026

- Li: Restriktionsenzyme
Plasmide & genom.
DNA
- Re: Oben: LAC
Unten: TAS



→ TAS2R38

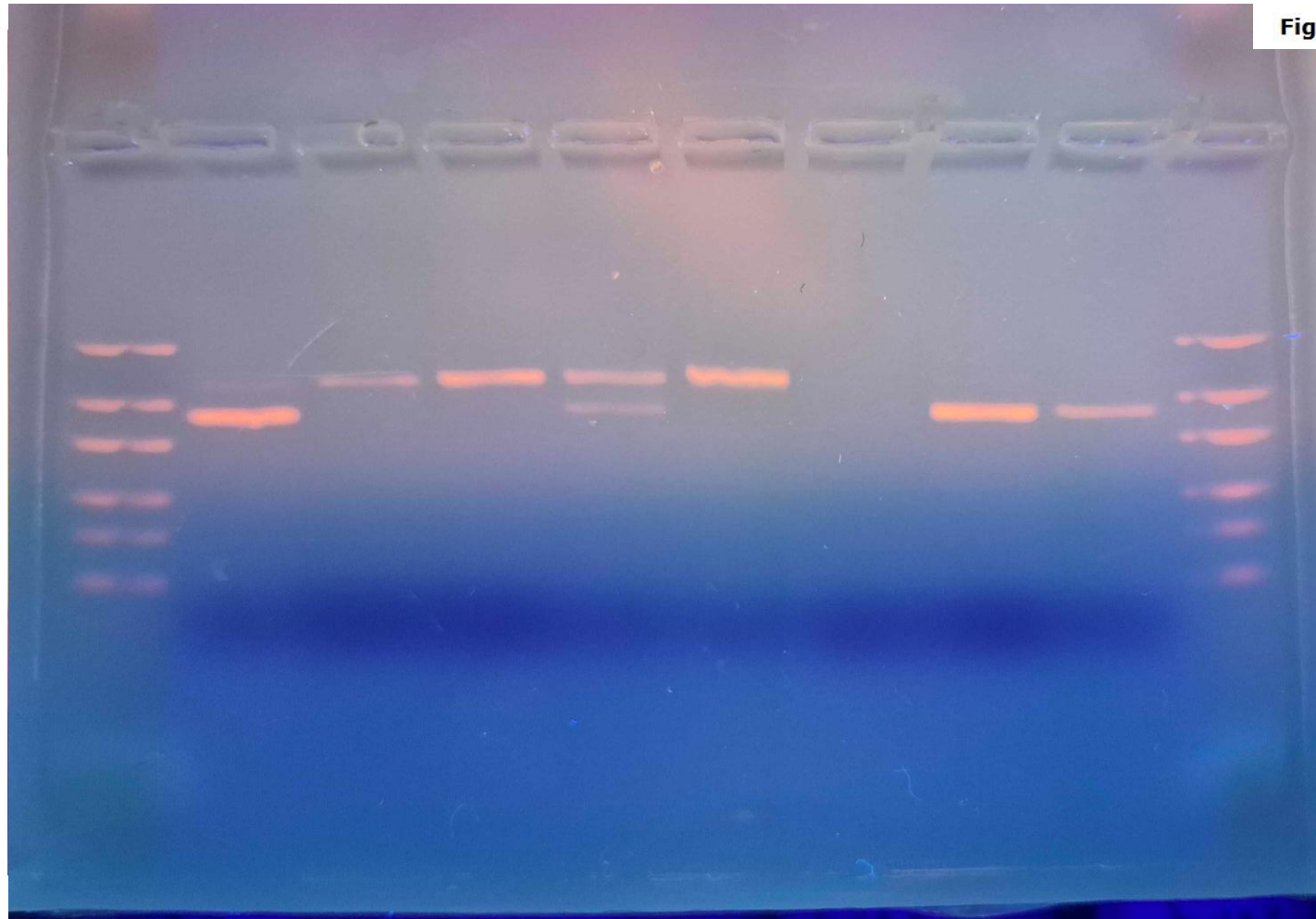
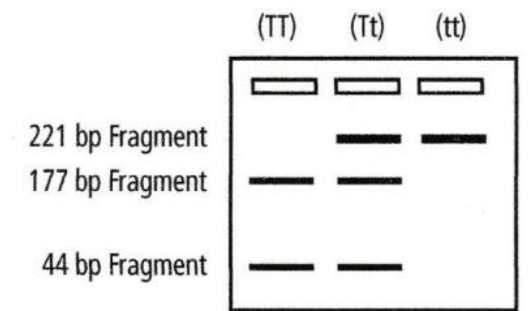
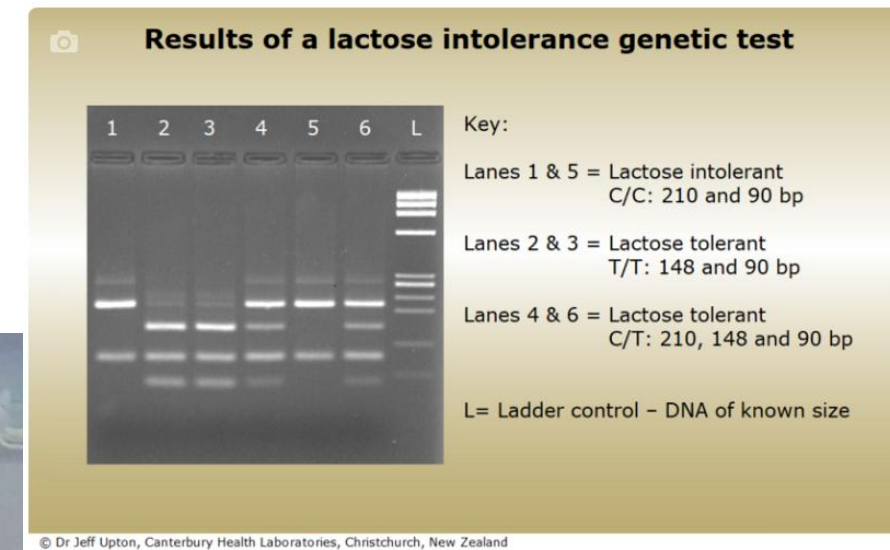
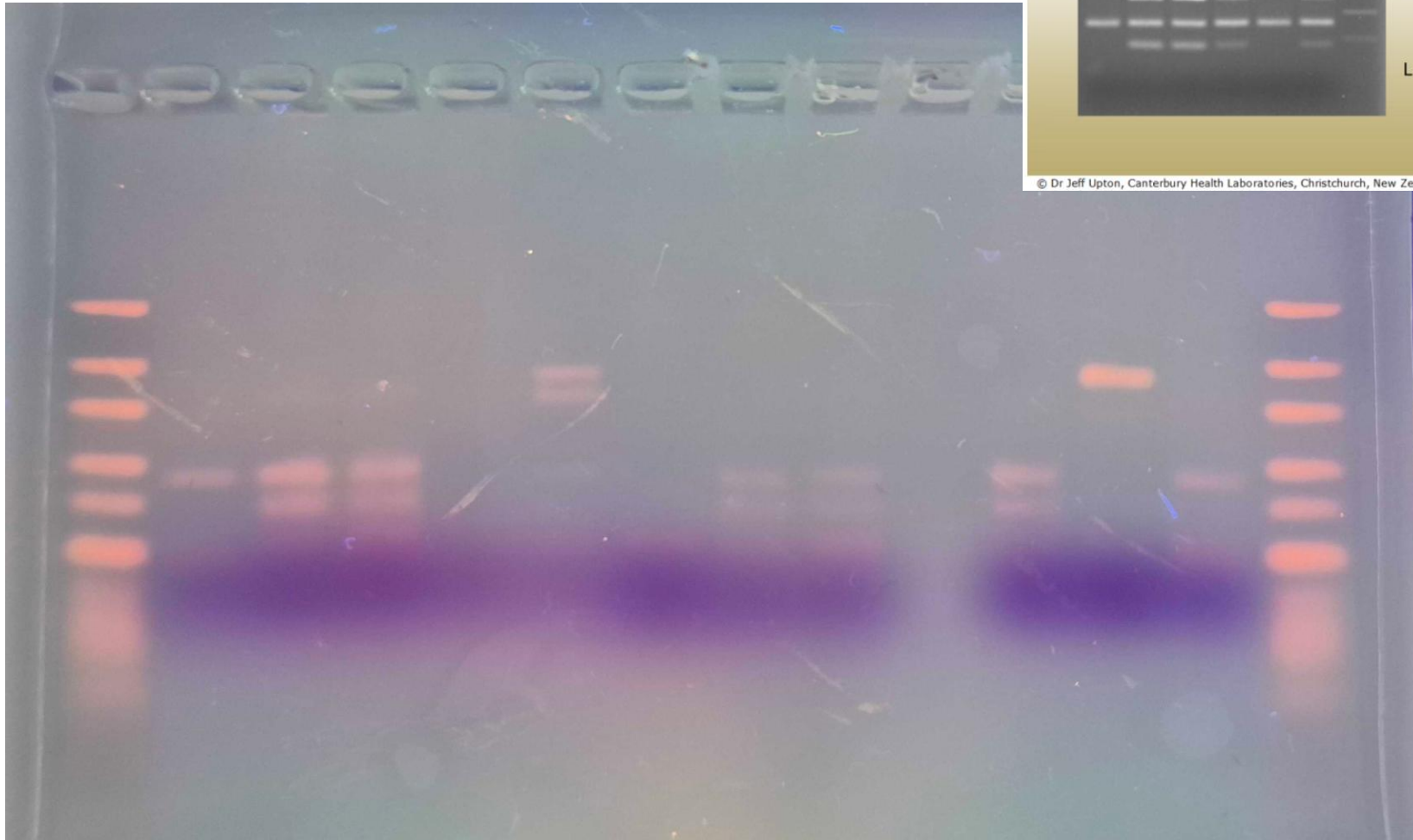


Figure 2. Analysis of genotypes of the PTC gene

→ LAC

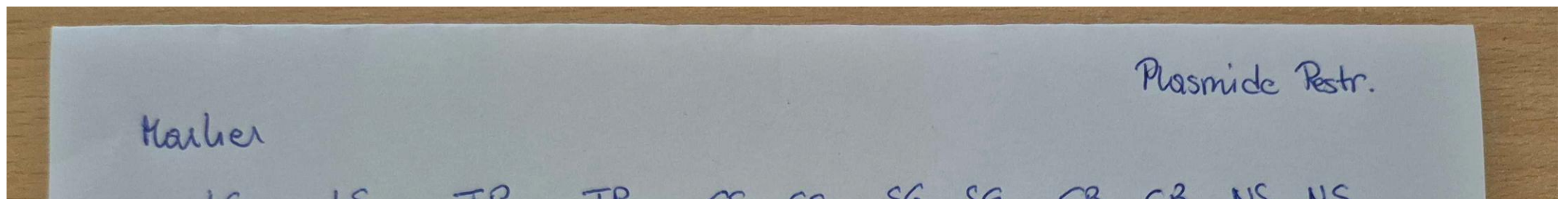
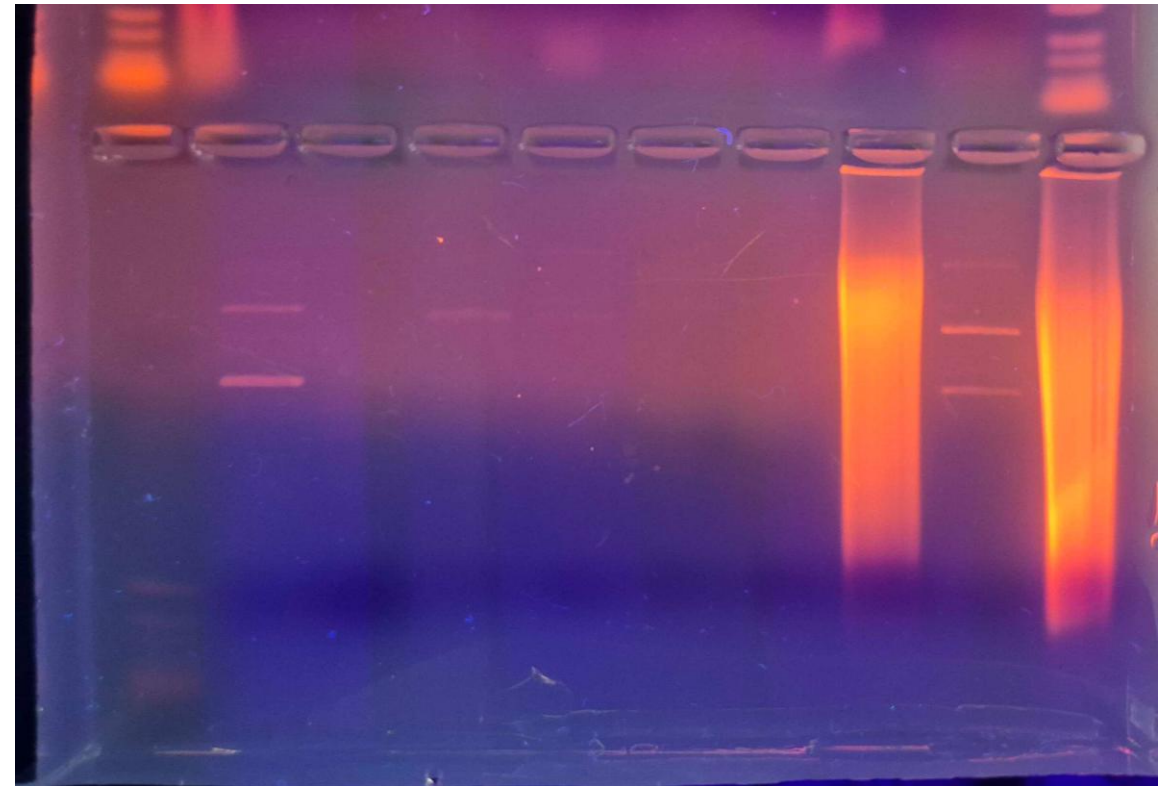
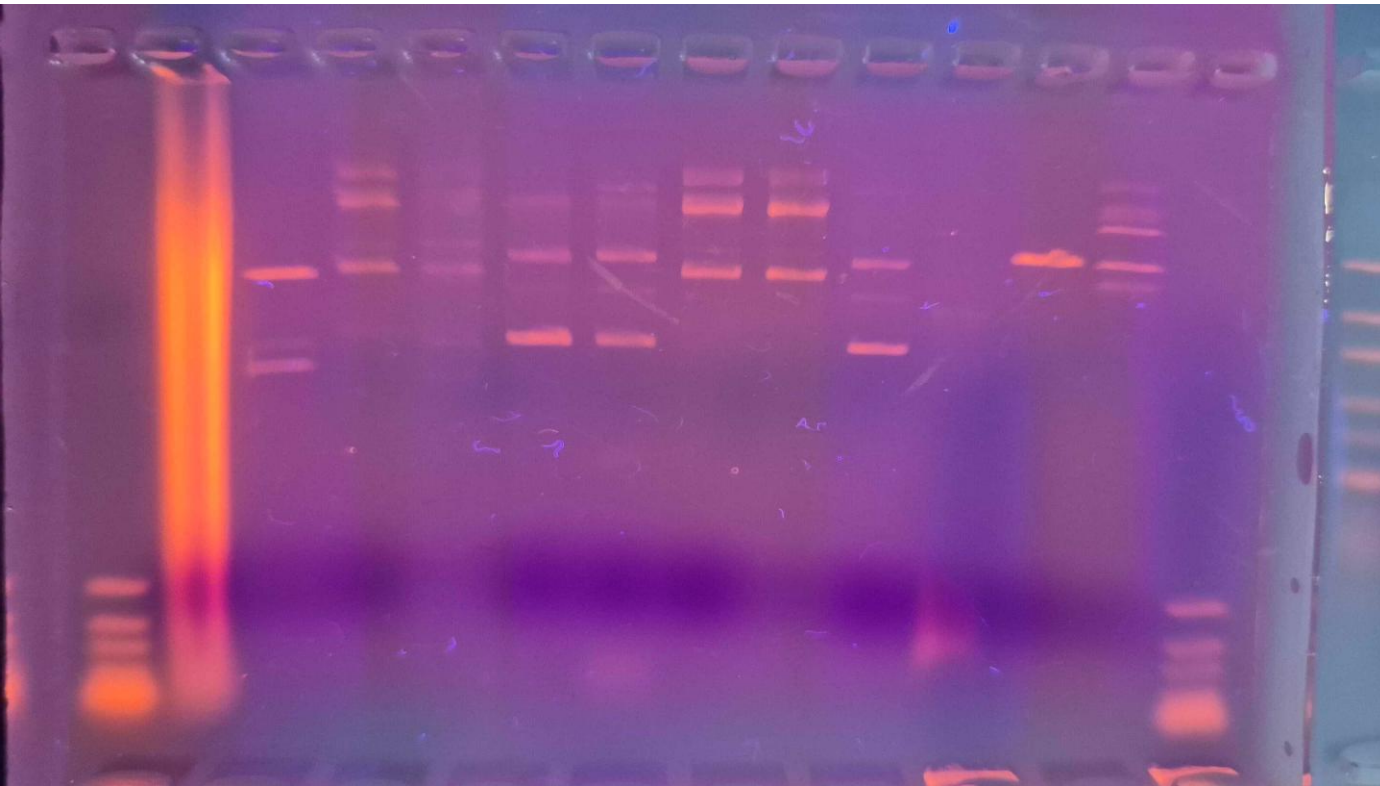


lac
 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | | AK MS JV ← LAC
 SS | SG | LS | TP | NS | CB | EL | AL |

Tas
 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
 SS | SG | LS | TP | NS | CB | EL | AL | ← TAS

28.05.2026

→ Plasmide & Genomische DNA (Restriktionsenzyme)



Einteilung Restriktionsenzyme (Plasmide & Genomische DNA)

1) Restriktionsenzyme

	alt	neu
①② EcoRI (HF)	①	②
③④ PstI	③	④
⑤ XmaI/XbaI	X	⑤
⑥ AluI	X	⑥
⑦ NaeI	X	⑦
⑧ HindIII	⑧	X
⑨ PvuII	⑨	X
⑩ BsuRI	⑩	X

a) 1 Enzym
17 μ L Plasmid (KIT) bzw. gen. DNA (mit RNase)
+ 2 μ L Puffer
+ 1 μ L Enzym

b) 2 Enzyme
16 μ L Plasmid (KIT)
+ 2 μ L Puffer
+ 2 x 1 μ L Enzym

→ mind. 1h im Wasserbad
↓
1% Agarosegel

2) RFLP
Restriktionsverdau
17 μ L PCR-Produkt
+ 2 μ L Puffer
+ 1 μ L Enzym
(1x TAS $\Rightarrow$ HaeIII
1x LAC $\Rightarrow$ RsaI)

5) ELISA

3) Protein / PAGE
20 μ L Probe
+ 20 μ L Loading Dye
→ 5' 95°C



BIOLABOR & I

WAS HAT INFORMATIK MIT BIO?

In Hagenberg sehr viel. Der Studiengang "Bioinformatik" bildet genau für diese Aufgaben aus. Ob zur Analyse von Krankheiten, bei der virtuellen 3D-Planung von Operationen oder bei der Berechnung von Proteinstrukturen. Erfahre mehr bei: www.fhnw.at

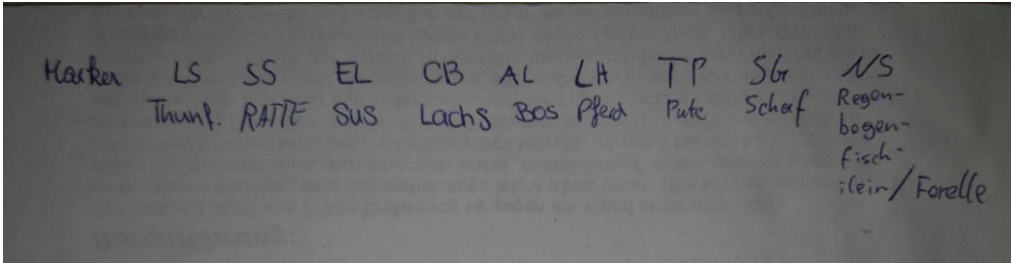
Proteine

Proteinbestimmung nach Bradford

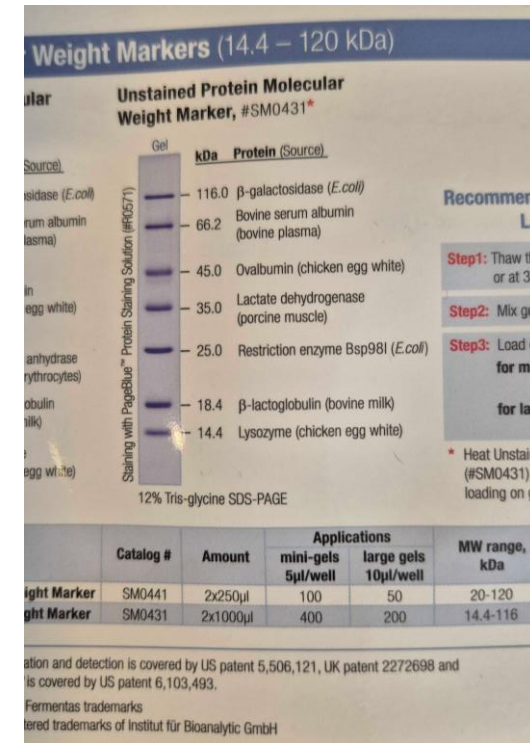
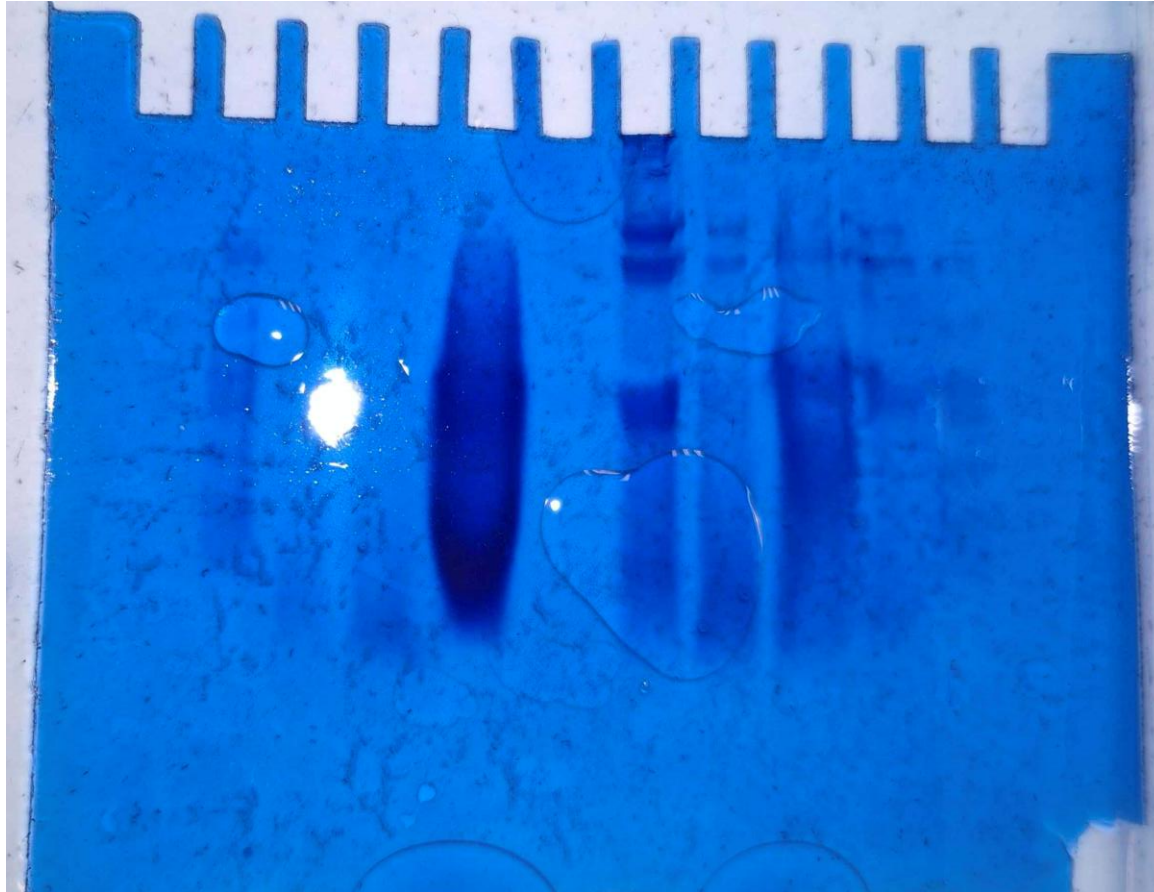
	4	2	1	0,5	0,25	0,125
20 µL Probe { 1) Eichg:	1,333	1,035	0,478	0,269	0,157	0,062
TP:	0,337					
NS(1/8):	0,228					
SG:	0,379 0,409					
2) Eichg:	0,883	0, 577 ⁴⁵⁷	0,245	0,110	0,040	0,025
SS 1/2:	0,194					
SB 1/10:	0,005					
Sus:	0,163					
3) Eichg:	0,898	0,411	0,111	0,23	0,000	-0,011
LS 1/2:	0,060					
AL:	0,149					
LH:	0,249					

Prokine

A photograph of a blue, textured surface, possibly a book cover or folder, with a white, irregular shape in the center. The surface is secured by several white, curved fasteners along the top edge. The background is a light beige color.



Proteine (2)



- 2 = Marker
- 11 = Marker

3	4	5	6	7	8	9	10
EL	AL	LS	CB	SS	SG	TP	NS
Sus	Bos	Thunfisch	Lachs	Ratte	Schaf	Pute	Forelle

HPLC

HPLC

HPLC		7.5.2026
1) Ctrl 10mg/mL 10mg/mL		1) Monster
2) Ctrl 1mg/mL		2) Schwed. Blaubeere Teekanne
3) Ctrl 0.1 mg/mL		3) Coke
4) Wildberry Schweppes Tonic		4) Espresso
5) Almdudler		5) Cola NS
6) Scharnerbombe & Cola		6) Kaffee TP
7) ?		7) Eiste Rauch Pfirsich JV
8) Monster 1:10	] ältere	
9) Cola (Pur)		
10) Cola (1:10)		

- Dateien siehe eLearning (Run 1- 3)